

Guide: The Correction-Coverage Gap Is Closed

Collatz proof project

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What Lean now proves

Suppose a positive natural Collatz orbit is unbounded. Then the sum of the reciprocals of all its successive values converges. This is a theorem about every hypothetical unbounded orbit, not about almost every starting seed.

The existing reciprocal theorem then gives bounded ideal correction. The previously unresolved case of an unbounded orbit with divergent reciprocal sum is now ruled out.

The key idea

An unbounded orbit never repeats a value. Its distinct values cannot merge after the same number of steps, since that would create a repetition. Most small starting values map into a smaller range after suitably many steps. Injectivity limits how many of them can belong to one unbounded orbit.

Lean proves that a finite set of S orbit values below 32^m satisfies $8^m S \leq 2 \cdot 216^m + 243^m$. Splitting the positive integers into shells from 32^m to 32^{m+1} turns this into a convergent reciprocal majorant.

What this adds to the proof program

Every hypothetical unbounded positive orbit now belongs to the existing bounded-correction framework. Its inverse drift $2^t/3^{j_t}$ tends to zero, and its shifted correction divided by its current value tends to zero. Neither statement supplies a fixed gap in average odd-step density.

The quantitative refinement gives an explicit budget $B_K \rightarrow 0$: an unbounded orbit that always stays above 32^K has reciprocal sum at most B_K and correction at most $N(e^{B_K} - 1)$. The technical paper gives the two geometric terms in B_K and their Lean proofs.

This bound does not make the absolute correction small. Lean proves that the upper envelope, even at $N = 32^K$, is at least $(1728/5)27^K$. That is a limitation of this estimate, not a lower bound on actual errors. A separate verified barrier puts actual shifted corrections above one. We therefore still need an arithmetic argument beyond rounding the ideal.

Why this is not Collatz yet

Bounded correction is a restriction on an unbounded orbit; it is not an exclusion of one. Nontrivial cycles also remain a separate unresolved case. Finite cycles repeat values, so their infinite reciprocal series diverges; they do not satisfy the new summability hypothesis.

The mathematical implication is related to a quantitative 1999 result of Garcia and Tal. We make no novelty claim. The main project's proof uses its own existing bounds and adds no external mathematical axiom.

Verification

The Lean build and 120-declaration axiom audit pass, as do three exact finite controls. Run `lake build Collatz.OrbitSummability` from `lean/`. The technical paper provides the argument and precise remaining gap.